05-410 UCRE Project Two Final Report

Team Breadcrumbs

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Executive Summary

Research Question
How might we use AI to reduce students' reliance on it for direct answers and increase its use as a step-by-step study tool that builds independent reasoning skills?

Activities Conducted

- Our team conducted comprehensive contextual research to understand the friction points in digital study workflows. Activities included:
- 12 In-Person Contextual Interviews focusing on recent, highly specific instances of AI usage for coursework.
- Interpretation Sessions to translate interview transcripts into single-idea interpretation notes.
- Affinity Diagramming to organically group interpretation notes and discover overarching themes regarding student emotions and AI workflows.
- Modeling using two different lenses
    - Customer Journey Map (focusing on emotional timelines)
    - Sequence/User Flow Model (focusing on system interactions and AI constraints).

What you learned/your high-level insights
We discovered that students do not instinctively default to AI to avoid work; they turn to it when traditional academic resources fail and time pressure mounts. Students explicitly desire to control how much help the AI gives them and often use it to expand on their own initial ideas. However, the over-delivering nature of current AI tools triggers emotional dissonance, in which students feel relieved that the task is done but guilty that their chance to learn was taken away.

Recommended next steps:
Based on these findings, the next phase of the project should focus on designing and testing an AI study tool that gives students more control over how help is delivered. Rather than defaulting to full answers, the tool should support adjustable scaffolding, such as hints, partial steps, or error checking, so that students can match the tool's support to their level of stress, time pressure, and learning goals. Future testing should evaluate whether this kind of controlled support helps students feel both more effective in the moment and more confident in their long-term understanding.

Research Methodology

Rationale for Selected Method
We utilized Contextual Interviews (in-person interviews about a past or recent experience) centered around a specific, recent memory of AI use. Because using AI for homework carries an academic stigma, a standard survey or abstract interview would likely yield sanitized, socially

desirable answers (faking good). By asking participants to physically walk us through a specific assignment in the environment they completed it (contextual recall), we bypassed the stigma and uncovered the triggers, such as late-night fatigue and specific moments of frustration, that drive AI adoption.

Summarize the research

Our team conducted a multi-phase generative qualitative research study. We executed 12 contextual interviews to gather raw observational data on student workflows. We then analyzed this data through structured interpretation sessions to translate transcripts into single-idea insights. Finally, we synthesized these insights using affinity diagramming to uncover broad behavioral themes, and built two structural models (a Customer Journey Map and an Ideal User Flow) to visualize the emotional and systemic breakdowns of current AI tools.

Participant Demographics and Recruitment

We conducted 12 in-depth contextual interviews (IC0 through IC11) with undergraduate students.

- Target Audience
    - We specifically screened for students who actively utilize GenAI tools to assist with their academic coursework.
- Demographics
    - Our participant pool represented a cross-section of academic disciplines, with a skew toward technical and STEM fields. Majors included Computer Science, Information Systems, Mechanical Engineering, Electrical Engineering, Physics, Biochemistry, and Psychology. This distribution allowed us to see how AI use cases shift between highly mechanical, logic-based coding tasks and writing- or theory-based assignments.
- Recruitment
    - Participants were recruited through a combination of personal campus networks across multiple universities (including CMU, UIUC, NYU, and Columbia).

Research Process, Form, and Goals

Each interview lasted between 15 and 30 minutes and followed a semi-structured Contextual Inquiry protocol. To ground the data in reality, participants were asked to recall a specific recent assignment in which they used GenAI. The interview was broken down into three core investigative phases

- Establishing Context of Use
    - We reconstructed the user's physical and digital environment. We probed for specific behavioral triggers, asking questions like, "Where were you working?",

"Did you have a split-screen setup?", and "What was the exact breaking point that made you open the AI?"

- Establishing Emotional Significance:
    - We mapped the user's emotional arc to understand the psychological drivers of reliance on AI. We asked them to describe their feelings before using the tool (e.g., frustration, panic), during the generation wait-time, and after submitting the assignment (e.g., relief vs. guilt).
- Connecting to Our Work
    - To test our hypotheses regarding AI as a tutor, we evaluated their actual prompting strategies (e.g., whether they asked for hints or full answers). We then presented hypothetical constraints and asked how they would react to an AI tool that explicitly refused to provide the final answer.
- Research Goals
    - Our primary objective was to observe the environmental triggers (time pressure, fatigue) and interface interactions that push a student toward the path of least resistance. This understanding is required to design an AI intervention that successfully balances helpful scaffolding with productive academic friction.

Insights

Supporting Evidence & Recommendations

Insight 1: Students do not turn to GenAI first because they want to avoid learning. They turn to it when traditional academic support breaks down and the cost of continuing alone becomes too high.

Across interviews, students described AI use as a response to a failure in their learning environment rather than a default preference for shortcuts. Many participants said they would ideally start with class materials, notes, or independent problem solving, and only use AI for clarification. However, when lectures were confusing, textbooks skipped steps, TAs were unavailable, or it was too late to get human help, AI became the only resource that felt responsive in the moment. This means the real problem is not simply "students rely on AI too much," but that existing support systems often fail at the exact moment students need help most. For our project, this suggests that a successful study-oriented AI should be designed less as a replacement for learning and more as an always-available bridge when institutional resources are too slow, unclear, or inaccessible.

Insight 2: The shift from learning mode to answer-seeking mode is usually caused by situational pressure, especially deadlines, fatigue, and cognitive overload.
A major pattern across the data is that students often begin with good intentions, but their behavior changes when they are overwhelmed. Participants described using AI differently when they were tired, stuck for too long, facing multiple deadlines, or trying to protect their sleep and mental health. In those moments, the goal stops being understanding and becomes finishing. This reveals that direct-answer seeking is not a stable trait, but a context-dependent coping strategy. This implies that if we want students to use AI in more productive ways, the tool has to account for fluctuating states of stress and urgency instead of assuming users are always motivated to learn step by step. A rigid "learning-only" system may fail if it does not acknowledge that students sometimes approach AI in survival mode.

Insight 3: Students experience emotional dissonance when using GenAI for coursework because the tool solves their immediate problem while also taking away the struggle they associate with real learning.
One of the clearest emotional patterns in the research was the coexistence of relief and guilt. Students felt grateful when AI helped them finish a task, find an error, or move on with their night, but many also felt uneasy because they believed they had skipped an important part of the learning process. Several participants worried about dependence, critical thinking loss, or not being prepared for closed-book exams later. This shows that students are not blindly trusting AI or using it without reflection. Instead, they are actively negotiating a tradeoff between short-term performance and long-term growth. That matters because it suggests there is real demand for a tool that helps students preserve a sense of intellectual ownership. Designing for emotional trust may be just as important as designing for correctness or speed.

Insight 4: Students want control over how much help AI gives them, but current tools are optimized to over-deliver rather than scaffold.
A repeated frustration in the data was not that AI was unhelpful, but that it was too helpful too quickly. Participants often wanted hints such as a first step, error checking, or a simplified explanation, yet current tools tended to jump to full solutions or skip the reasoning path the student was trying to preserve. Some students explicitly described wanting a TA-like interaction or a system with adjustable levels of help, such as showing only the first line of code, then the logic, then the full solution if needed. This points to a fundamental mismatch between current LLM design and student learning needs: standard chatbots are optimized for task completion, while students in academic settings often need controlled scaffolding. For our project, this means the opportunity is not just to make AI "more educational," but to give users nuanced control over the relationship between pacing and answer reveal.

Insight 5: Students already have a fairly sophisticated idea of what "good" AI-supported learning looks like, which means the gap is not awareness but tool fit.
The interviews suggest that most students already understand the difference between using AI to

think and using AI to bypass thinking. Many described ideal behaviors such as attempting the problem first, asking for clarification instead of answers, using AI to generate practice questions, or wanting explanations tied to their own class materials. In other words, students do not need to be taught from scratch that step-by-step support is better for learning. They already know that. The issue is that current tools make the less productive path easier, faster, and more tempting, especially under stress. This reframes our design challenge: instead of trying to convince students to value learning, the better opportunity is to build a system that makes their preferred learning behavior easier to sustain in real contexts.

Appendix

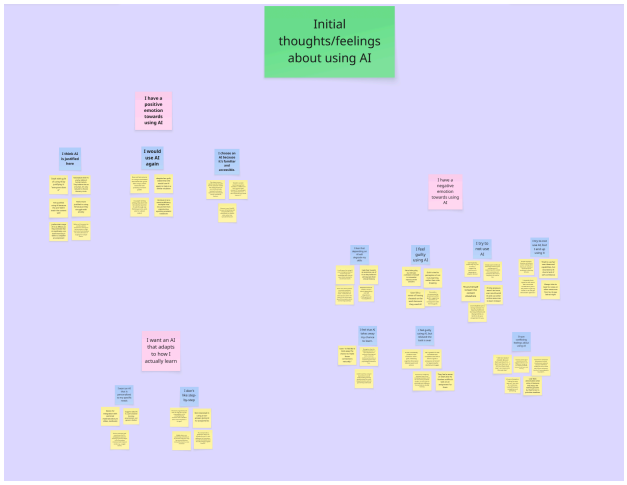
Interview Interpretation Notes

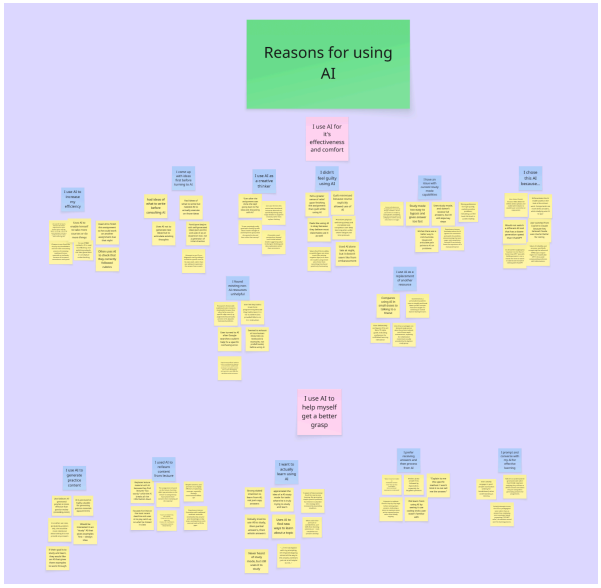
Interpretation Notes Spreadsheet link: Interpretation notes (breadcrumbs)

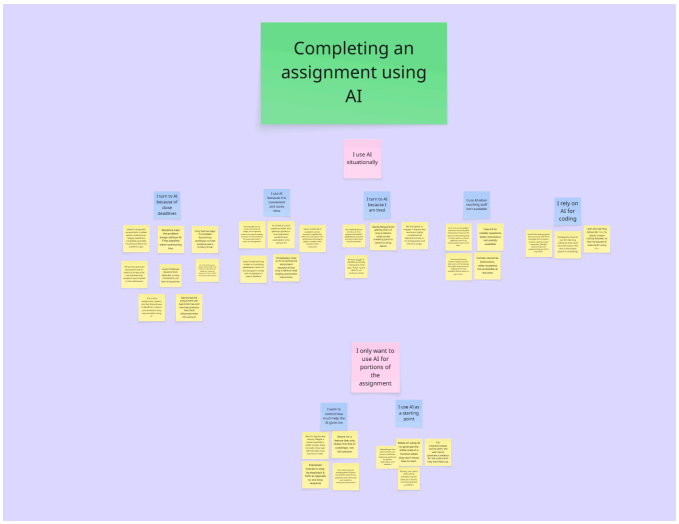
Affinity Diagramming

Link to the Miro board: [Link]

Screenshots of the Affinity Diagramming









Modeling

Journey Mapping

Student

Scenario: A student who earnestly tries at first, faces time pressure, uses AI ineffectively, and feels both relief and guilt upon completing the assignment

	Assignment Created	Struggle/Confusion	Time Pressure	AI Use	Post Reflection
Doing	Looking at the assignment reading the details	Trying to work on the assignment legitimately Maybe using AI to help learn	Panicking because of due date Stressed, worried about grade	Using AI to get answers Plugging worksheet or questions straight into AI	Working on other assignments Guilty of previous assignments
Thinking	**"I should try to learn this"**	**"I don't really understand / I've gotten stuck"**	**"The deadline is soon, and I just need to finish and submit"**	**"This is so much faster"** **"This might be bad for my learning"**	**"I didn't really learn that thoroughly"** **"But at least it is done"**
Feeling	They feel.. **Neutral, Motivated**	**I feel frustrated /anxious that I do not understand this**	**Spike in Stress; I am worried about getting a bad grade**	**I already feel relieved getting this out of the way**	**They feel... Relieved, Guilty**

User Flow

